Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20191001-027-R0	Issued	2019-10-18

UT Thickness Re-baseline Survey - ATU-100

Scope and Responsibilities

This report details the re-baseline ultrasonic thickness survey conducted at the Hamilton Plant for the ATU-100. The scope included the measurement of 150 Condition Monitoring Locations (CMLs) across 18 piping circuits to establish a new data foundation following the adoption of QSCP-CIS-001 Rev 0. The survey was performed by Trevor Nolan and Julia Martinez, both qualified to ASNT SNT-TC-1A, ensuring compliance with corporate data requirements and RBI-based interval settings.

Larkhall NDE Services Inc. was responsible for the execution of the NDT activities, while David Park provided technical oversight and quality management for the project. The inspection focused on critical circuits including sour gas, rich amine, and regenerator overhead lines. All activities were coordinated to align with the updated requirements for CUI monitoring and profile radiography as mandated by the revised site integrity standards.

Inspection Procedures and Standards

Measurements were performed under QSCP-NDT-001 Rev 2, which incorporates specific requirements for insulation removal and the use of inspection ports. The survey adhered to the technical guidelines of ASTM E797 and API 570 for piping inspection. The Olympus 37DL PLUS 0611452 was utilized for all readings, with its accuracy verified against LNDE-CAL-2019-0588 prior to the commencement of site work.

For the stainless steel reflux lines 3-RF-101 and 2-RF-102, the ultrasonic gauge was adjusted to the appropriate material velocity setting. The design parameters for the piping systems, including those for 6-SG-101 and 8-TG-101, were referenced from ASME B31.3 to ensure the re-baseline data correctly reflects the system's mechanical limits and minimum allowable wall thicknesses.

Reference Documentation and History

The current survey strategy cites QSCP-IP-ATU100 Rev 3, which rationalized the CML count to 150 locations. This plan supersedes the previous full-unit survey documented in LNDE-RPT-20160118-004-R0. The transition to a 3-year RBI interval for rich amine and regenerator overhead lines was influenced by the damage mechanism reviews and historical data captured during the first Larkhall NDE Services Inc. engagement in LNDE-RPT-20140902-024-R0.

Historical readings for lines such as 4-RA-101, 4-RA-102, and 6-RA-103 were compared against the new re-baseline values to calculate current corrosion rates. The data integration process ensures that all Class 1 and Class 2 systems, including the lean amine supply lines 4-LA-101 and 4-LA-102, meet the rigorous data requirements established by the QSCP Corporate Asset Integrity department.

Survey Findings and Analysis

The survey covered the primary process lines 6-SG-102, 6-LA-103, and the L/R exchanger circuits 4-LA-104 and 4-RA-104. Utility lines 2-ST-101, 3-CW-101, and 3-CW-102 were also measured to confirm baseline integrity. Most circuits showed wall thicknesses well above t-min, however, specific locations on the reflux line 2-RF-102 exhibited accelerated thinning. CML 2-RF-102-06 recorded a wall of 2.74 mm, resulting in a remaining life of 2.9 years.

Other notable readings on 2-RF-102 include CML 04 at 2.83 mm, CML 05 at 2.88 mm, and CML 07 at 2.82 mm, with remaining lives ranging from 3.2 to 4.5 years. These findings on the regenerator overhead and reflux systems, including 3-RV-101 and 2-RV-102, necessitate strict adherence to the 3-year inspection interval. No readings were found below the retirement thickness for the sour gas inlet lines or the treated gas line 8-TG-101.

Thickness readings, measured 2019-10-18

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2016-02-12	6.17	5.93	0.050	0.065	36.3	2022-10-18	
6-SG-101-02	7.11	3.56	2016-02-12	5.76	5.51	0.065	0.068	28.7	2022-10-18	
6-SG-101-03	7.11	3.56	2016-02-12	5.87	5.56	0.064	0.084	23.7	2022-10-18	
6-SG-101-04	7.11	3.56	2016-02-12	5.58	5.34	0.074	0.065	24.0	2022-10-18	
6-SG-101-05	7.11	3.56	2016-02-12	5.38	5.04	0.089	0.092	16.0	2022-10-18	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-06	7.11	3.56	2016-02-12	5.88	5.56	0.064	0.087	23.0	2022-10-18	
6-SG-101-07	7.11	3.56	—	—	6.24	—	—	—	2022-10-18	
6-SG-101-08	7.11	3.56	2016-02-12	6.00	5.77	0.065	0.063	34.2	2022-10-18	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2016-02-12	5.42	5.05	0.089	0.101	14.8	2022-10-18	
6-SG-102-02	7.11	3.56	2016-02-12	5.43	5.10	0.085	0.090	17.2	2022-10-18	
6-SG-102-03	7.11	3.56	2016-02-12	5.22	4.92	0.090	0.082	15.1	2022-10-18	
6-SG-102-04	7.11	3.56	2016-02-12	5.18	4.90	0.089	0.076	15.1	2022-10-18	
6-SG-102-05	7.11	3.56	2016-02-12	6.19	6.07	0.045	0.033	>50	2022-10-18	
6-SG-102-06	7.11	3.56	2016-02-12	6.06	6.03	0.044	0.008	>50	2022-10-18	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2016-02-12	7.76	7.63	0.024	0.035	>50	2025-10-18	
8-TG-101-02	8.18	4.06	2016-02-12	7.46	7.33	0.037	0.035	>50	2025-10-18	
8-TG-101-03	8.18	4.06	2016-02-12	7.54	7.37	0.036	0.046	>50	2025-10-18	
8-TG-101-04	8.18	4.06	2016-02-12	7.51	7.38	0.030	0.035	>50	2025-10-18	
8-TG-101-05	8.18	4.06	2016-02-12	7.73	7.57	0.028	0.043	>50	2025-10-18	
8-TG-101-06	8.18	4.06	—	—	7.75	—	—	—	2025-10-18	
8-TG-101-07	8.18	4.06	2016-02-12	7.39	7.17	0.060	0.060	>50	2025-10-18	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2016-02-12	5.02	4.70	0.055	0.087	17.5	2025-10-18	
4-LA-101-02	6.02	3.18	2016-02-12	5.45	5.26	0.032	0.052	40.3	2025-10-18	
4-LA-101-03	6.02	3.18	2016-02-12	5.45	5.38	0.025	0.019	>50	2025-10-18	
4-LA-101-04	6.02	3.18	2016-02-12	5.48	5.29	0.033	0.052	40.9	2025-10-18	
4-LA-101-05	6.02	3.18	2016-02-12	5.54	5.38	0.027	0.043	>50	2025-10-18	
4-LA-101-06	6.02	3.18	2016-02-12	5.30	5.22	0.036	0.022	>50	2025-10-18	
4-LA-101-07	6.02	3.18	2016-02-12	4.87	4.86	0.048	0.003	35.1	2025-10-18	
4-LA-101-08	6.02	3.18	—	—	5.04	—	—	—	2025-10-18	
4-LA-101-09	6.02	3.18	2016-02-12	5.00	4.86	0.038	0.038	44.2	2025-10-18	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2016-02-12	5.25	5.09	0.039	0.043	43.9	2025-10-18	
4-LA-102-02	6.02	3.18	2016-02-12	5.36	5.15	0.039	0.057	34.5	2025-10-18	
4-LA-102-03	6.02	3.18	2016-02-12	5.28	5.15	0.039	0.035	>50	2025-10-18	
4-LA-102-04	6.02	3.18	2016-02-12	4.90	4.65	0.057	0.068	21.6	2025-10-18	
4-LA-102-05	6.02	3.18	2016-02-12	5.34	5.23	0.033	0.030	>50	2025-10-18	
4-LA-102-06	6.02	3.18	2016-02-12	5.60	5.52	0.020	0.022	>50	2025-10-18	
4-LA-102-07	6.02	3.18	2016-02-12	5.40	5.32	0.029	0.022	>50	2025-10-18	
4-LA-102-08	6.02	3.18	2016-02-12	5.70	5.55	0.041	0.041	>50	2025-10-18	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2016-02-12	6.44	6.40	0.026	0.011	>50	2025-10-18	
6-LA-103-02	7.11	3.56	2016-02-12	6.53	6.49	0.030	0.011	>50	2025-10-18	
6-LA-103-03	7.11	3.56	2016-02-12	6.28	6.23	0.036	0.014	>50	2025-10-18	
6-LA-103-04	7.11	3.56	2016-02-12	6.15	5.85	0.056	0.082	28.1	2025-10-18	
6-LA-103-05	7.11	3.56	2016-02-12	5.98	5.81	0.058	0.046	38.5	2025-10-18	
6-LA-103-06	7.11	3.56	2016-02-12	5.82	5.46	0.072	0.098	19.4	2025-10-18	
6-LA-103-07	7.11	3.56	2016-02-12	6.12	5.93	0.053	0.052	44.8	2025-10-18	
6-LA-103-08	7.11	3.56	2016-02-12	6.49	6.19	0.036	0.082	32.3	2025-10-18	
6-LA-103-09	7.11	3.56	—	—	6.16	—	—	—	2025-10-18	
6-LA-103-10	7.11	3.56	2016-02-12	6.30	6.00	0.082	0.082	29.9	2025-10-18	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2016-02-12	7.83	7.29	0.052	0.147	23.7	2022-10-18	
4-RA-101-02	8.56	3.81	2016-02-12	7.94	7.39	0.049	0.149	24.0	2022-10-18	
4-RA-101-03	8.56	3.81	2016-02-12	7.65	6.64	0.083	0.274	10.3	2022-10-18	
4-RA-101-04	8.56	3.81	2016-02-12	8.00	7.54	0.046	0.125	29.8	2022-10-18	
4-RA-101-05	8.56	3.81	2016-02-12	7.82	7.18	0.060	0.174	19.4	2022-10-18	
4-RA-101-06	8.56	3.81	2016-02-12	7.87	6.94	0.191	0.253	12.4	2022-10-18	
4-RA-101-07	8.56	3.81	2016-02-12	8.14	7.68	0.039	0.125	31.0	2022-10-18	
4-RA-101-08	8.56	3.81	2016-02-12	7.31	6.45	0.094	0.234	11.3	2022-10-18	
4-RA-101-09	8.56	3.81	2016-02-12	8.13	7.68	0.038	0.122	31.6	2022-10-18	
4-RA-101-10	8.56	3.81	2016-02-12	8.02	7.55	0.047	0.128	29.3	2022-10-18	
4-RA-101-11	8.56	3.81	2016-02-12	7.60	6.66	0.086	0.255	11.2	2022-10-18	
4-RA-101-12	8.56	3.81	—	—	7.03	—	—	—	2022-10-18	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2016-02-12	8.02	7.58	0.041	0.120	31.5	2022-10-18	
4-RA-102-02	8.56	3.81	2016-02-12	7.64	6.96	0.068	0.185	17.0	2022-10-18	
4-RA-102-03	8.56	3.81	2016-02-12	7.55	7.05	0.064	0.136	23.8	2022-10-18	
4-RA-102-04	8.56	3.81	2016-02-12	7.72	7.04	0.064	0.185	17.5	2022-10-18	
4-RA-102-05	8.56	3.81	2016-02-12	7.67	7.06	0.063	0.166	19.6	2022-10-18	
4-RA-102-06	8.56	3.81	2016-02-12	7.73	7.15	0.058	0.158	21.2	2022-10-18	
4-RA-102-07	8.56	3.81	2016-02-12	7.78	7.30	0.054	0.130	26.8	2022-10-18	
4-RA-102-08	8.56	3.81	2016-02-12	7.35	6.40	0.089	0.258	10.0	2022-10-18	
4-RA-102-09	8.56	3.81	2016-02-12	7.54	6.78	0.075	0.207	14.4	2022-10-18	
4-RA-102-10	8.56	3.81	—	—	7.36	—	—	—	2022-10-18	
4-RA-102-11	8.56	3.81	2016-02-12	7.96	7.72	0.054	0.065	>50	2022-10-18	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2016-02-12	10.29	9.83	0.050	0.125	42.1	2022-10-18	
6-RA-103-02	10.97	4.57	2016-02-12	9.69	8.61	0.105	0.294	13.8	2022-10-18	
6-RA-103-03	10.97	4.57	2016-02-12	10.06	9.49	0.064	0.155	31.8	2022-10-18	
6-RA-103-04	10.97	4.57	2016-02-12	9.31	8.22	0.117	0.296	12.3	2022-10-18	
6-RA-103-05	10.97	4.57	2016-02-12	9.48	8.33	0.111	0.313	12.0	2022-10-18	
6-RA-103-06	10.97	4.57	2016-02-12	10.32	9.79	0.052	0.144	36.2	2022-10-18	
6-RA-103-07	10.97	4.57	2016-02-12	10.20	9.59	0.059	0.166	30.3	2022-10-18	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-08	10.97	4.57	2016-02-12	10.33	9.73	0.053	0.163	31.6	2022-10-18	
6-RA-103-09	10.97	4.57	2016-02-12	10.33	9.78	0.052	0.149	34.9	2022-10-18	
6-RA-103-10	10.97	4.57	2016-02-12	9.49	8.37	0.112	0.304	12.5	2022-10-18	
6-RA-103-11	10.97	4.57	2016-02-12	9.77	8.91	0.234	0.234	18.5	2022-10-18	
6-RA-103-12	10.97	4.57	—	—	9.52	—	—	—	2022-10-18	
6-RA-103-13	10.97	4.57	2016-02-12	10.15	9.55	0.104	0.163	30.5	2022-10-18	
6-RA-103-14	10.97	4.57	—	—	8.26	—	—	—	2022-10-18	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2016-02-12	6.34	6.05	0.065	0.079	31.6	2022-10-18	
3-RV-101-02	7.62	3.56	2016-02-12	6.31	5.97	0.068	0.092	26.1	2022-10-18	
3-RV-101-03	7.62	3.56	2016-02-12	5.63	5.24	0.097	0.106	15.9	2022-10-18	
3-RV-101-04	7.62	3.56	2016-02-12	5.88	5.58	0.085	0.082	23.7	2022-10-18	
3-RV-101-05	7.62	3.56	2016-02-12	6.36	6.11	0.066	0.068	37.5	2022-10-18	
3-RV-101-06	7.62	3.56	2016-02-12	5.96	5.55	0.086	0.111	17.9	2022-10-18	
3-RV-101-07	7.62	3.56	2016-02-12	5.54	5.07	0.124	0.128	11.8	2022-10-18	
3-RV-101-08	7.62	3.56	2016-02-12	6.31	5.86	0.093	0.122	18.8	2022-10-18	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2016-02-12	4.35	4.16	0.058	0.052	23.6	2022-10-18	
2-RV-102-02	5.54	2.79	2016-02-12	4.01	3.86	0.073	0.041	14.6	2022-10-18	
2-RV-102-03	5.54	2.79	2016-02-12	4.64	4.37	0.047	0.073	21.5	2022-10-18	
2-RV-102-04	5.54	2.79	2016-02-12	4.73	4.59	0.042	0.038	42.9	2022-10-18	
2-RV-102-05	5.54	2.79	2016-02-12	4.97	4.78	0.029	0.052	38.5	2022-10-18	
2-RV-102-06	5.54	2.79	2016-02-12	4.27	4.27	0.060	0.000	24.5	2022-10-18	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2016-02-12	5.31	5.40	0.003	-0.024	>50	2025-10-18	
3-RF-101-02	5.49	2.41	2016-02-12	5.34	5.38	0.003	-0.011	>50	2025-10-18	
3-RF-101-03	5.49	2.41	2016-02-12	5.33	5.35	0.004	-0.005	>50	2025-10-18	
3-RF-101-04	5.49	2.41	2016-02-12	5.35	5.29	0.007	0.016	>50	2025-10-18	
3-RF-101-05	5.49	2.41	2016-02-12	5.33	5.17	0.012	0.043	>50	2025-10-18	
3-RF-101-06	5.49	2.41	2016-02-12	5.23	5.12	0.017	0.030	>50	2025-10-18	
3-RF-101-07	5.49	2.41	2016-02-12	5.37	5.38	0.002	-0.003	>50	2025-10-18	
3-RF-101-08	5.49	2.41	—	—	5.23	—	—	—	2025-10-18	
3-RF-101-09	5.49	2.41	2016-02-12	5.19	5.13	-0.002	0.016	>50	2025-10-18	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2016-02-12	3.76	3.66	0.011	0.027	>50	2025-10-18	
2-RF-102-02	3.91	2.16	2016-02-12	3.64	3.65	0.010	-0.003	>50	2025-10-18	
2-RF-102-03	3.91	2.16	2016-02-12	3.75	3.68	0.011	0.019	>50	2025-10-18	
2-RF-102-04	3.91	2.16	2016-02-12	3.38	2.83	0.044	0.149	4.5	2022-01-14	RL/2 < interval
2-RF-102-05	3.91	2.16	2016-02-12	3.49	2.88	0.133	0.166	4.3	2021-12-19	RL/2 < interval
2-RF-102-06	3.91	2.16	2016-02-12	3.47	2.74	0.159	0.198	2.9	2021-04-04	RL/2 < interval
2-RF-102-07	3.91	2.16	2016-02-12	3.57	2.82	0.048	0.204	3.2	2021-06-01	RL/2 < interval

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2016-02-12	5.31	5.16	0.039	0.041	48.6	2025-10-18	
4-LA-104-02	6.02	3.18	2016-02-12	5.00	4.79	0.053	0.057	28.2	2025-10-18	
4-LA-104-03	6.02	3.18	2016-02-12	5.42	5.23	0.031	0.052	39.7	2025-10-18	
4-LA-104-04	6.02	3.18	2016-02-12	5.63	5.49	0.024	0.038	>50	2025-10-18	
4-LA-104-05	6.02	3.18	2016-02-12	4.88	4.71	0.056	0.046	27.2	2025-10-18	
4-LA-104-06	6.02	3.18	2016-02-12	5.55	5.38	0.030	0.046	47.6	2025-10-18	
4-LA-104-07	6.02	3.18	2016-02-12	5.28	5.15	0.036	0.035	>50	2025-10-18	
4-LA-104-08	6.02	3.18	2016-02-12	5.11	4.88	0.045	0.063	27.2	2025-10-18	
4-LA-104-09	6.02	3.18	—	—	4.61	—	—	—	2025-10-18	
4-LA-104-10	6.02	3.18	2016-02-12	4.84	4.63	0.057	0.057	25.4	2025-10-18	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2016-02-12	7.71	7.15	0.058	0.152	21.9	2022-10-18	
4-RA-104-02	8.56	3.81	2016-02-12	7.85	7.31	0.053	0.147	23.8	2022-10-18	
4-RA-104-03	8.56	3.81	2016-02-12	8.28	7.94	0.071	0.092	44.7	2022-10-18	
4-RA-104-04	8.56	3.81	2016-02-12	8.18	7.64	0.097	0.147	26.1	2022-10-18	
4-RA-104-05	8.56	3.81	2016-02-12	8.16	7.57	0.095	0.160	23.5	2022-10-18	
4-RA-104-06	8.56	3.81	2016-02-12	7.61	7.13	0.062	0.130	25.5	2022-10-18	
4-RA-104-07	8.56	3.81	2016-02-12	7.32	6.67	0.081	0.177	16.2	2022-10-18	
4-RA-104-08	8.56	3.81	2016-02-12	7.81	7.46	0.083	0.095	38.4	2022-10-18	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2016-02-12	3.61	3.59	0.014	0.005	>50	2025-10-18	
2-ST-101-02	3.91	2.29	2016-02-12	3.55	3.34	0.024	0.057	18.4	2025-10-18	
2-ST-101-03	3.91	2.29	2016-02-12	3.55	3.68	0.011	-0.035	>50	2025-10-18	
2-ST-101-04	3.91	2.29	2016-02-12	3.68	3.52	0.013	0.043	28.3	2025-10-18	
2-ST-101-05	3.91	2.29	2016-02-12	3.50	3.27	0.027	0.063	15.7	2025-10-18	
2-ST-101-06	3.91	2.29	2016-02-12	3.63	3.66	-0.008	-0.008	>50	2025-10-18	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2016-02-12	5.03	4.91	0.022	0.033	>50	2025-10-18	
3-CW-101-02	5.49	2.79	2016-02-12	5.01	4.98	0.018	0.008	>50	2025-10-18	
3-CW-101-03	5.49	2.79	2016-02-12	4.93	4.87	0.028	0.016	>50	2025-10-18	
3-CW-101-04	5.49	2.79	2016-02-12	5.26	5.20	0.011	0.016	>50	2025-10-18	
3-CW-101-05	5.49	2.79	2016-02-12	5.35	5.14	0.014	0.057	41.2	2025-10-18	
3-CW-101-06	5.49	2.79	2016-02-12	5.08	5.01	0.019	0.019	>50	2025-10-18	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2016-02-12	5.06	4.85	0.028	0.057	36.1	2025-10-18	
3-CW-102-02	5.49	2.79	2016-02-12	4.37	4.17	0.057	0.054	24.3	2025-10-18	
3-CW-102-03	5.49	2.79	2016-02-12	4.54	4.27	0.053	0.073	20.2	2025-10-18	
3-CW-102-04	5.49	2.79	2016-02-12	4.69	4.50	0.043	0.052	33.1	2025-10-18	
3-CW-102-05	5.49	2.79	2016-02-12	4.66	4.45	0.057	0.057	29.1	2025-10-18	

Sign-off	Name	Function	Date	Signature
Prepared	Trevor Nolan	UT technician, SNT-TC-1A UT Level II	2019-10-18	_____
Prepared	Julia Martinez	UT/RT technician, SNT-TC-1A UT/RT Level II	2019-10-18	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2019-10-18	_____